GPTZero

INVESTIGATIONS FEATURED

GPTZero finds 100 new hallucinations in NeurIPS 2025 accepted papers

GPTZero's analysis 4841 papers accepted by NeurIPS 2025 show there are at least 100 with confirmed hallucinations

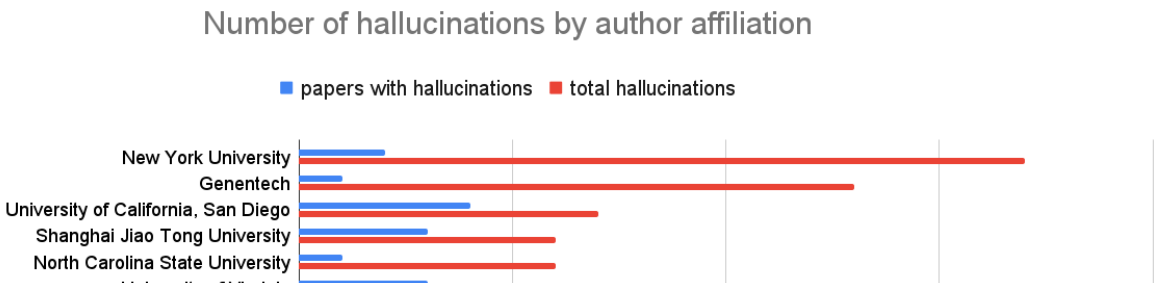




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NEURAL INFORMATION PROCESSING SYSTEMS

The Conference on Neural Information Processing Systems (NeurIPS) is one of the most prestigious AI conferences in the world. The most recent meeting occurred in November 2025, where 100+ hallucinated citations were published. Last month, GPTZero used our Hallucination Check tool to uncover 50 hallucinated citations in papers under review for ICLR 2026. However, we knew that ICLR (the International Conference on Learning Representations) was just one of hundreds of academic conferences and publications besieged by a tsunami of AI slop. After scanning 4841 papers accepted by the equally prestigious Conference on Neural Information Processing Systems (NeurIPS), we discovered 100s of hallucinated citations missed by the 3+ reviewers who evaluated each paper. **Below, we uncover 100 confirmed hallucinations in the table below, spanning over 51 NeurIPS papers,** which were not previously reported.



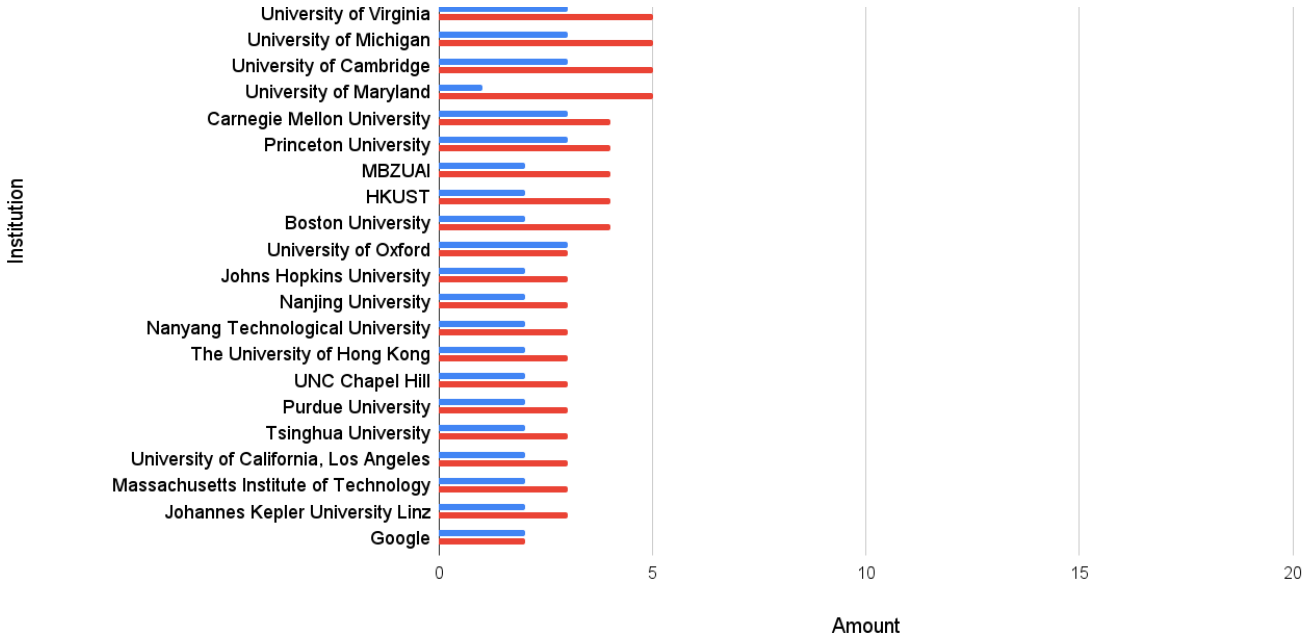


Figure 1: Distribution of hallucinations by author's institution. How we computed this: a paper with 2 hallucinations with any authors from University A and University B will count as 2 hallucinations and 1 paper with hallucinations for both universities, independent of the number of authors from either university.

A Problem of Scale

ICLR, NeurIPS, ICML, and AAAI are the top machine learning / artificial intelligence conferences in the world, drawing thousands of submissions and participants annually. However, a submission tsunami fueled by generative AI, paper mills, and publication pressure has strained these conferences' review pipelines to the breaking point. Between 2020 and 2025, submissions to NeurIPS increased more than 220% from 9,467 to 21,575. In response, organizers have had to recruit ever greater numbers of reviewers, resulting in issues of oversight, expertise alignment, negligence, and even fraud.

Our purpose in publishing these results is to illuminate a critical vulnerability in the peer review pipeline, not criticize the specific organizers, area chairs, or reviewers who participated in NeurIPS 2025. Over the past several years NeurIPS has changed the review process several times to address problems created by submission volume and generative AI tools. Still, our results reveal the consequences of a system that leaves academic reviewers, editors, and conference organizers outnumbered and outgunned — trying to protect the rigor of peer review against challenges it was never designed to defend against.

Table of 100 Hallucinated Citations in Published Across 53 NeurIPS Papers

These NeurIPS papers have already been accepted, presented live, and effectively published. Since NeurIPS 2025 had an acceptance rate for main track papers of 24.52%, each of these papers beat out 15,000 other papers despite containing one or more hallucinations. This is concerning, given that the NeurIPS LLM policy considers hallucinated citations to be grounds for a paper's rejection or revocation, similar to ICLR.

*We've scanned each paper for both hallucinated citations (Sources) and AI-generated text (AI). An "** next to the scan indicates the paper is likely a mix of AI and human text, while "**" indicates the paper is likely AI-generated.*

Published Paper	GPTZero Scan	Example of Verified Hallucination	Comment
SimWorld: An Open-ended Simulator for Agents in Physical and Social Worlds	Sources AI	John Doe and Jane Smith. Webvoyager: Building an end-to-end web agent with large multimodal models. arXiv preprint	Article with a matching title exists here. Authors are obviously fabricated. arXiv ID links to a

		arXiv:2401.00001, 2024.	different article.
Unmasking Puppeteers: Leveraging Biometric Leakage to Expose Impersonation in AI-Based Videoconferencing	Sources AI*	John Smith and Jane Doe. Deep learning techniques for avatar-based interaction in virtual environments. IEEE Transactions on Neural Networks and Learning Systems, 32(12):5600-5612, 2021. doi: 10.1109/TNNLS.2021.3071234. URL https://ieeexplore.ieee.org/document/307123	No author or title match. Doesn't exist in publication. URL and DOI are fake.
Unmasking Puppeteers: Leveraging Biometric Leakage to Expose Impersonation in AI-Based Videoconferencing	Sources AI*	Min-Jun Lee and Soo-Young Kim. Generative adversarial networks for hyper-realistic avatar creation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 1234-1243, 2022. doi: 10.1109/CVPR.2022.001234. URL https://ieeexplore.ieee.org/ document/00123	No author or title match. Doesn't exist in publication. URL and DOI are fake.
SimWorld-Robotics: Synthesizing Photorealistic and Dynamic Urban Environments for Multimodal Robot Navigation and Collaboration	Sources AI*	Firstname Lastname and Others. Drivlme: A large-scale multi-agent driving benchmark, 2023. URL or arXiv ID to be updated.	No title or author match. Potentially referring to this article, but year is off (2024)
SimWorld-Robotics: Synthesizing Photorealistic and Dynamic Urban Environments for Multimodal Robot Navigation and Collaboration	Sources AI*	Firstname Lastname and Others. Robotslang: Grounded natural language for multi-robot object search, 2024. To appear.	No title or author match. Potentially referring to this article, but year is totally off (2020).
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Nuo Lou and et al. Dsp: Diffusion-based span prediction for masked text modeling. arXiv preprint arXiv:2305.XXXX, 2023.	No title or author match and arXiv ID is incomplete.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	A. Sahoo and et al. inatk: Iterative noise aware text denoising. arXiv preprint arXiv:2402.XXXX, 2024.	No title or author match and arXiv ID is incomplete.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Sheng Shi and et al. Maskgpt: Uniform denoising diffusion for language. arXiv preprint arXiv:2401.XXXX, 2024.	No title or author match and arXiv ID is incomplete.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Asma Issa, George Mohler, and John Johnson. Paraphrase identification using deep contextualized representations. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 517-526, 2018.	No author or title match. No match in publication.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Yi Tay, Kelvin Fu, Kai Wu, Ivan Casanueva, Jianfeng Liu, Byron Wallace, Shuohang Wang, Bajrang Singh, and Julian McAuley. Reasoning with heterogeneous graph representations for knowledge-aware question answering. In Findings of the Association for Computational Linguistics: ACL 2021, pp. 3497-3506, 2021.	No exact author or title match, although this title is close. No match in the publication.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Alex Wang, Rishi Bommasani, Dan Hendrycks, Daniel Song, and Zhilin Zhang. Efficient fewshot learning with efl: A single transformer for all tasks. In arXiv preprint arXiv:2107.13586, 2021.	No title or author match. ArXiv ID leads to a different article.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Lei Yu, Jimmy Dumsmyr, and Kevin Knight. Deep paraphrase identification. In Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. $650-655,2014$.	No title or author match. No match in publication
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	X. Ou and et al. Tuqdm: Token unmasking with quantized diffusion models. In ACL, 2024.	No title or author match.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Franz Aichberger, Lily Chen, and John Smith. Semantically diverse language generation. In International Conference on Learning Representations (ICLR), 2025.	No title or author match. Some similarity to this article
Efficient semantic uncertainty quantification in language models via	Sources	Maria Glushkova, Shiori Kobayashi, and Junichi Suzuki. Uncertainty estimation in neural text regression. In Findings of the	No author or title match. No match in publication

language models via diversity-steered sampling	AI	Association for Computational Linguistics: EMNLP 2021, pp. $4567-4576,2021$.	in publication.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Yichao Wang, Bowen Zhou, Adam Lopez, and Benjamin Snyder. Uncertainty quantification in abstractive summarization. In Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 1234-1245, 2022.	No author or title match.
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Mohit Jain, Ethan Perez, and James Glass. Learning to predict confidence for language models. In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 245-256, 2021.	No author or title match. No match in publication
Efficient semantic uncertainty quantification in language models via diversity-steered sampling	Sources AI	Srinivasan Kadavath, Urvashi Khandelwal, Alec Radford, and Noam Shazeer. Answer me this: Self-verifying large language models. In arXiv preprint arXiv:2205.05407, 2022.	No author or title match. ArXiv ID leads to a different article.
Privacy Reasoning in Ambiguous Contexts	Sources AI	Zayne Sprague, Xi Ye, Kyle Richardson, and Greg Durrett. MuSR: Testing the limits of chain-of-thought with multistep soft reasoning. In EMNLP, 2023.	Two authors are omitted and one (Kyle Richardson) is added. This paper was published at ICLR 2024.
Memory-Augmented Potential Field Theory: A Framework for Adaptive Control in Non-Convex Domains	Sources AI**	Mario Paolone, Trevor Gaunt, Xavier Guillaud, Marco Liserre, Sakis Meliopoulos, Antonello Monti, Thierry Van Cutsem, Vijay Vittal, and Costas Vournas. A benchmark model for power system stability controls. IEEE Transactions on Power Systems, 35(5):3627-3635, 2020.	The authors match this paper, but the title, publisher, volume, issue, and page numbers are incorrect. Year (2020) is correct.
Memory-Augmented Potential Field Theory: A Framework for Adaptive Control in Non-Convex Domains	Sources AI**	Mingliang Han, Bingni W Wei, Phelan Senatus, Jörg D Winkel, Mason Youngblood, I-Han Lee, and David J Mandell. Deep koopman operator: A model-free approach to nonlinear dynamical systems. Chaos: An Interdisciplinary Journal of Nonlinear Science, 30(12):123135, 2020.	No title or author match. Journal and other identifiers match this article.
Adaptive Quantization in Generative Flow Networks for Probabilistic Sequential Prediction	Sources AI	Francisco Ramalho, Meng Liu, Zihan Liu, and Etienne Mathieu. Towards gflownets for continuous control. arXiv preprint arXiv:2310.18664, 2023.	No author or title match. ArXiv ID matches this paper.
Grounded Reinforcement Learning for Visual Reasoning	Sources AI*	Arjun Gupta, Xi Victoria Lin, Chunyuan Zhang, Michel Galley, Jianfeng Gao, and Carlos Guestrin Ferrer. Robust compositional visual reasoning via language-guided neural module networks. In Advances in Neural Information Processing Systems (NeurIPS), 2021.	No title or author match. This paper has a similar title and matches publication.
MTRec: Learning to Align with User Preferences via Mental Reward Models	Sources AI	Diederik P. Kingma and Jimmy Ba. Deepfm: a factorization-machine based neural network for ctr prediction. In International Conference on Learning Representations, 2015.	Title matches this paper. Authors, date, and publisher match this paper.
Redefining Experts: Interpretable Decomposition of Language Models for Toxicity Mitigation	Sources AI*	Weijia Xu, Xing Niu, and Marine Carpuat. Controlling toxicity in neural machine translation. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 4245-4256, 2020.	Authors, publisher and date match this paper. Title and page numbers don't match.
Redefining Experts: Interpretable Decomposition of Language Models for Toxicity Mitigation	Sources AI*	Xiang Zhang, Xuehai Wei, Xian Zhang, and Xue Zhang. Adversarial attacks and defenses in toxicity detection: A survey. In Proceedings of the 2020 International Joint Conference on Neural Networks (IJCNN), pages 1-8. IEEE, 2020.	No author or title match. Doesn't exist in publication.
Self-supervised Learning of Echocardiographic Video Representations via Online Cluster Distillation	Sources AI*	Fenglin Ding, Debesh Jha, Maria Härgestam, Pål Halvorsen, Michael A Riegler, Dag Johansen, Ronny Hänsch, and Håvard Stensland. Vits: Vision transformer for video self-supervised pretraining of surgical phase recognition. In International Conference on Medical Image Computing and Computer-Assisted Intervention, pages 293-302. Springer, 2022.	No title or author match. Proceedings from this conference are split into volumes, but the citation doesn't have a volume number.
PANTHER: Generative		Humberto Acevedo-Viloria, Juan Martinez, and Maria Garcia. Relational graph	

Pretraining Beyond Language for Sequential User Behavior Modeling	Sources AI*	convolutional networks for financial fraud detection. IEEE Transactions on Knowledge and Data Engineering, 33(7):1357-1370, 2021. doi: 10.1109/TKDE.2020.3007655.	No author or title match. Doesn't exist in the cited publication.
PANTHER: Generative Pretraining Beyond Language for Sequential User Behavior Modeling	Sources AI*	Majid Zolghadr, Mohsen Jamali, and Jiawei Zhang. Diffurecsys: Diffusion-based generative modeling for sequential recommendation. Proceedings of the ACM Web Conference (WWW), pages 2156-2165, 2024. doi: 10.1145/3545678.3557899.	No author or title match. DOI doesn't exist.
LiteReality: Graphic-Ready 3D Scene Reconstruction from RGB-D Scans	Sources AI	Bernd Kerbl, Thomas Müller, and Paolo Favaro. Efficient 3d gaussian splatting for real-time neural rendering. In CVPR, 2022. 2, 3	Loosely matches this article, but only one author and part of the title actually match.
LiteReality: Graphic-Ready 3D Scene Reconstruction from RGB-D Scans	Sources AI	Punchana Khungurn, Edward H. Adelson, Julie Dorsey, and Holly Rushmeier. Matching real-world material appearance. TPAMI, 2015. 6	No clear match. Two authors and the subject match this article.
When and How Unlabeled Data Provably Improve In-Context Learning	Sources AI	Ashish Kumar, Logan Engstrom, Andrew Ilyas, and Dimitris Tsipras. Understanding self-training for gradient-boosted trees. In Advances in Neural Information Processing Systems (NeurIPS), volume 33, pp. 1651-1662, 2020.	No title or author match. Doesn't exist in publication.
When and How Unlabeled Data Provably Improve In-Context Learning	Sources AI	Chuang Fan, Shipeng Liu, Seyed Motamed, Shiyu Zhong, Silvio Savarese, Juan Carlos Niebles, Anima Anandkumar, Adrien Gaidon, and Stefan Scherer. Expectation maximization pseudo labels. arXiv preprint arXiv:2305.01747, 2023.	This paper exists, but all the authors are fabricated.
DualCnst: Enhancing Zero-Shot Out-of-Distribution Detection via Text-Image Consistency in Vision-Language Models	Sources AI*	T. Qiao, W. Liu, Z. Xie, H. Xu, J. Lin, J. Huang, and Y. Yang, "Clip-score: A robust scoring metric for text-to-image generation," arXiv preprint arXiv:2201.07519, 2022.	No clear author or title matches. Title loosely matches this article. ArXiv ID leads here.
Optimal Rates for Generalization of Gradient Descent for Deep ReLU Classification	Sources AI	Yunwen Lei, Puyu Wang, Yiming Ying, and Ding-Xuan Zhou. Optimization and generalization of gradient descent for shallow relu networks with minimal width. preprint, 2024.	No title match. Authors match this paper.
GeoDynamics: A Geometric State-Space Neural Network for Understanding Brain Dynamics on Riemannian Manifolds	Sources AI*	Uher, R., Goodman, R., Moutoussis, M., Brammer, M., Williams, S.C.R., Dolan, R.J.: Cognitive and neural predictors of response to cognitive behavioral therapy for depression: a review of the evidence. Journal of Affective Disorders 169, 94-104 (2014)	No exact title or author match. Loose title match with this article. Doesn't exist in the journal volume
Robust Label Proportions Learning	Scan AI*	Junyeong Lee, Yiseong Kim, Seungju Park, and Hyunjik Lee. Softmatch: Addressing the quantity-quality trade-off in semi-supervised learning. In Advances in Neural Information Processing Systems (NeurIPS), volume 36, pages 18315-18327, 2023.	Title matches this paper. No match in NeurIPS volume 36.
NFL-BA: Near-Field Light Bundle Adjustment for SLAM in Dynamic Lighting	Sources AI	Z. Zhu, T. Yu, X. Zhang, J. Li, Y. Zhang, and Y. Fu. Neuralrgb-d: Neural representations for depth estimation and scene mapping. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2022.	No author or title match. Doesn't exist in publication.
NFL-BA: Near-Field Light Bundle Adjustment for SLAM in Dynamic Lighting	Sources AI	Y. Zhang, M. Oswald, and D. Cremers. Airslam: Illumination-invariant hybrid slam. In International Conference on Computer Vision (ICCV), pages 2345-2354, 2023.	No author or title match. Doesn't exist in publication.
Geometric Imbalance in Semi-Supervised Node Classification	Sources AI	Yihong Zhu, Junxian Li, Xianfeng Han, Shirui Pan, Liang Yao, and Chengqi Wang. Spectral contrastive graph clustering. In International Conference on Learning Representations, 2022.	No title or author match. This paper has a similar title, but there's no match in the ICLR 2022 database.
Geometric Imbalance in Semi-Supervised Node Classification	Sources AI	Ming Zhong, Han Liu, Weizhu Zhang, Houyu Wang, Xiang Li, Maosong Sun, and Xu Han. Hyperbolic and spherical embeddings for long-tail entities. In ACL, pages 5491-5501, 2021.	No author or title match. Doesn't exist publication.
NUTS: Eddy-Robust	Sources	Ye Gao, Robert Tardif, Jiale Cao, and Tapio Schneider. Artificial intelligence reconstructs	Title and publisher match this

Reconstruction of Surface Ocean Nutrients via Two-Scale Modeling	Sources AI*	Schneider. Artificial intelligence reconstructs missing climate information. Nature Geoscience, 17:158-164, 2024. doi: 10.1038/s41561-023-01297-2.	article. Issue, page numbers, and year match this article. DOI is fabricated.
NUTS: Eddy-Robust Reconstruction of Surface Ocean Nutrients via Two-Scale Modeling	Sources AI*	Étienne Pardoux and Alexander Yu Veretennikov. Poisson equation for multiscale diffusions. Journal of Mathematical Sciences, 111(3):3713-3719, 2002.	Authors have frequently published together on the "poisson equation", but this title doesn't match any of their publications. Doesn't exist in publication volume/issue.
Test-Time Adaptation of Vision-Language Models for Open-Vocabulary Semantic Segmentation	Sources AI*	Charanpal D Mummadi, Matthias Arens, and Thomas Brox. Test-time adaptation for continual semantic segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 11828-11837, 2021.	No title or author match. Doesn't exist in publication.
Test-Time Adaptation of Vision-Language Models for Open-Vocabulary Semantic Segmentation	Sources AI*	Jiacheng He, Zhilu Zhang, Zhen Wang, and Yan Huang. Autoencoder based test-time adaptation for semantic segmentation. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pages 998-1007, 2021.	No author or title match. Doesn't exist in publication.
Global Minimizers of ℓp-Regularized Objectives Yield the Sparsest ReLU Neural Networks	Sources AI	M. Gong, F. Yu, J. Zhang, and D. Tao. Efficient ℓ_{p} norm regularization for learning sparsity in deep neural networks. IEEE Transactions on Neural Networks and Learning Systems, 33(10): $5381-5392,2022	No title or author match.
SHAP Meets Tensor Networks: Provably Tractable Explanations with Parallelism	Sources AI	Mihail Stoian, Richard Milbradt, and Christian Mendl. NP-Hardness of Optimal TensorNetwork Contraction and Polynomial-Time Algorithms for Tree Tensor Networks. Quantum, 6:e119, 2022.	The authors match this article and the title is similar. However, the year, publisher and other data don't match. This article didn't appear in the 2022 Quantum volume.
SHAP Meets Tensor Networks: Provably Tractable Explanations with Parallelism	Sources AI	Jianyu Xu, Wei Li, and Ming Zhao. Complexity of Optimal Tensor Network Contraction Sequences. Journal of Computational Physics, 480:112237, 2023.	No title or author match. Doesn't exist in publication.
Learning World Models for Interactive Video Generation	Sources AI	Patrick Esser, Robin Rombach, and Björn Ommer. Structure-aware video generation with latent diffusion models. arXiv preprint arXiv:2303.07332, 2023.	Authors match this article. ArXiv ID leads to a different article.
Fourier Token Merging: Understanding and Capitalizing Frequency Domain for Efficient Image Generation	Sources AI	Lele Xu, Chen Lin, Hongyu Zhao, and et al. Gaborvit: Global attention with local frequency awareness. In European Conference on Computer Vision (ECCV), 2022.	No author or title match. No match in publication.
Fourier Token Merging: Understanding and Capitalizing Frequency Domain for Efficient Image Generation	Sources AI	Yoonwoo Lee, Jaehyeong Kang, Namil Kim, Jinwoo Shin, and Honglak Lee. Structured fast fourier transform attention for vision transformers. In Advances in Neural Information Processing Systems (NeurIPS), 2022.	No author or title match. Doesn't exist in publication.
Fourier Token Merging: Understanding and Capitalizing Frequency Domain for Efficient Image Generation	Sources AI	Siyuan Gong, Alan Yu, Xiaohan Chen, Yinpeng Lin, and Larry S Davis. Vision transformer compression: Early exiting and token pruning. In Advances in Neural Information Processing Systems (NeurIPS), 2021.	No author or title match. No match in publication.
Fourier Token Merging: Understanding and Capitalizing Frequency Domain for Efficient Image Generation	Sources AI	Jiuxiang Shi, Zuxuan Wu, and Dahua Lin. Token-aware adaptive sampling for efficient diffusion models. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.	No author or title match. Doesn't exist in publication.
Fourier Token Merging: Understanding and Capitalizing Frequency Domain for Efficient Image Generation	Sources AI	Raphael Muller, Simon Kornblith, and Geoffrey Hinton. Adavit: Adaptive tokens for efficient vision transformer. In Proceedings of the International Conference on Machine Learning (ICML), 2021.	Authors match this article. Title matches this article. No match in publication.
Fourier Token Merging: Understanding and Capitalizing Frequency Domain for Efficient Image Generation	Sources AI	Xin Wang, Anlin Chen, Lihui Xie, Xin Jin, Cheng Wang, and Ping Luo. Not all tokens are equal: Efficient transformer for tokenization and beyond. In Advances in Neural Information Processing Systems (NeurIPS), 2021.	No author or title match. This article title is similar. No match in publication.
A Unified Stability Analysis of SAM vs SGD: Role of Data	Sources	Z. Chen and N. Flammarion. When and why sam generalizes better: An optimization	No author or title match. ArXiv ID

Coherence and Emergence of Simplicity Bias	AI	perspective. arXiv preprint arXiv:2206.09267, 2022.	leads to a different paper.
A Unified Stability Analysis of SAM vs SGD: Role of Data Coherence and Emergence of Simplicity Bias	Sources AI	K. A. Sankararaman, S. Sankararaman, H. Pandey, S. Ganguli, and F. Bromberg. The impact of neural network overparameterization on gradient confusion and stochastic gradient descent. In 37th International Conference on Machine Learning (ICML), pages 8469-8479, 2020.	This paper is a match, but all authors but the first (K. A. Sankararaman) are fabricated.
MaterialRefGS: Reflective Gaussian Splatting with Multi-view Consistent Material Inference	Sources AI	Why Physically-Based Rendering. Physically-based rendering. Procedia IUTAM, 13(127137):3, 2015 .	No author given and title appears to be garbled. Publisher, issue, year, and pages match this article.
Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets	Sources AI	Pierre Casgrain, Anirudh Kulkarni, and Nicholas Watters. Learning to trade with continuous action spaces: Application to market making. arXiv preprint arXiv:2303.08603, 2023.	No title or author match. ArXiv ID matches a different article.
Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets	Sources AI	Z Ning and Y K Kwok. Q-learning for option pricing and hedging with transaction costs. Applied Economics, 52(55):6033-6048, 2020.	No author or title match. No match in journal volume/issue.
Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets	Sources AI	W L Chan and R O Shelton. Can machine learning improve delta hedging? Journal of Derivatives, $9(1): 39-56,2001$.	No author or title match. No match in journal volume/issue.
Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets	Sources AI	Petter N Kolm, Sebastian Krügel, and Sergiy V Zadorozhnyi. Reinforcement learning for optimal hedging. The Journal of Trading, 14(4):4-17, 2019.	No author or title match. There is no volume 14 of this journal.
Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets	Sources AI	Kyung Hyun Park, Hyeong Jin Kim, and Woo Chang Kim. Deep reinforcement learning for limit order book-based market making. Expert Systems with Applications, 169:114338, 2021.	No author or title match. Publisher ID matches this article.
FlowMixer: A Depth-Agnostic Neural Architecture for Interpretable Spatiotemporal Forecasting	Sources AI	Moonseop Han and Elizabeth Qian. Robust prediction of dynamical systems with structured neural networks: Long-term behavior and chaos. Physica D: Nonlinear Phenomena, 427:133006, 2021.	No author or title match. Publisher ID matches this article.
FlowMixer: A Depth-Agnostic Neural Architecture for Interpretable Spatiotemporal Forecasting	Sources AI	Bart De Schutter and Serge P Hoogendoorn. Modeling and control of freeway traffic flow by state space neural networks. Neural Computing and Applications, 17(2):175-185, 2008.	No title match, although Schutter and Hoogendorn have written or coauthored several related papers (example and example). Journal volume/issue matches an unrelated article.
FlowMixer: A Depth-Agnostic Neural Architecture for Interpretable Spatiotemporal Forecasting	Sources AI	Jaideep Pathak, Brian R Hunt, Georg M Goerg, and Themistoklis P Sapsis. Data-driven prediction of chaotic dynamics: Methods, challenges, and opportunities. Annual Review of Condensed Matter Physics, 14:379-401, 2023.	No author or title match. No match in journal volume.
FlowMixer: A Depth-Agnostic Neural Architecture for Interpretable Spatiotemporal Forecasting	Sources AI	Alejandro Güemes, Stefano Discetti, and Andrea Ianiro. Coarse-grained physics-based prediction of three-dimensional unsteady flows via neural networks. Science Advances, 7(46):eabj0751, 2021.	No title or author match. Doesn't exist in journal volume/issue.
BNMusic: Blending Environmental Noises into Personalized Music	Sources AI*	Jeongseung Park, Minseon Yang, Minz Won Park, and Geonseok Lee. Diffsound: Differential sound manipulation with a few-shot supervision. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pages 1767-1775, 2021.	No title or author match. Doesn't exist in publication.
Towards Multiscale Graph-based Protein Learning with Geometric Secondary Structural Motifs	Sources AI*	Wenxuan Sun, Tri Dao, Hongyu Zhuang, Zihang Dai, Albert Gu, and Christopher D Manning. Llamba: Efficient llms with mamba-based distillation. arXiv preprint arXiv:2502.14458, 2024.	ArXiv ID leads to this article with a similar title and one matching author.
Towards Multiscale Graph-based Protein Learning with Geometric Secondary Structural Motifs	Sources AI*	Tri Dao, Shizhe Ma, Wenxuan Sun, Albert Gu, Sam Smith, Aapo Kyrola, Christopher D Manning, and Christopher Re. An empirical study of state space models for large language modeling. arXiv preprint	Two authors (Tri Dao and Albert Gu), the arXiv ID, and the year match this paper. However, the title is only a partial match.

		arXiv:2406.07887, 2024.	
Fourier Clouds: Fast Bias Correction for Imbalanced Semi-Supervised Learning	Sources AI	Junyan Zhu, Chenyang Li, Chao He, and et al. Freematch: A simple framework for long-tailed semi-supervised learning. In NeurIPS, 2021.	No author or title match. This paper title is very close, but it was published by ICLR 2023 not NeurIPS 2021.
NormFit: A Lightweight Solution for Few-Shot Federated Learning with Non-IID Data	Scan AI	Yijie Zang et al. Fedclip: A federated learning framework for vision-language models. In NeurIPS, 2023.	No author or title match, although this title is close. No match in publication.
AI-Generated Video Detection via Perceptual Straightening	Sources AI	Jiahui Liu and et al. Tall-swin: Thumbnail layout transformer for generalised deepfake video detection. In ICCV, 2023.	No author or title match. A paper with a similar title appears in publication.
Multi-Expert Distributionally Robust Optimization for Out-of-Distribution Generalization	Sources AI*	Nitish Srivastava and Ruslan R Salakhutdinov. Discriminative features for fast frame-based phoneme classification. Neural networks, 47:17-23, 2013.	No title match, but authors have published together previously (example). No match in publication.
MIP against Agent: Malicious Image Patches Hijacking Multimodal OS Agents	Sources AI	Anh Tuan Nguyen, Shengping Li, and Chao Qin. Multimodal adversarial robustness: Attack and defense. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2021.	No author or title match. Doesn't exist in publication.
ACT as Human: Multimodal Large Language Model Data Annotation with Critical Thinking	Sources AI*	Jack Lau, Ankan Gayen, Philipp Tschandl, Gregory A Burns, Jiahong Yuan, Tanveer SyedaMahmood, and Mehdi Moradi. A dataset and exploration of models for understanding radiology images through dialogue. In Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 2575-2584, 2018.	No author match. Title matches another hallucinated citation in this paper. Doesn't exist in publication.
OCTDiff: Bridged Diffusion Model for Portable OCT Super-Resolution and Enhancement	Sources AI	Yikai Zhang et al. "Text-to-Image Diffusion Models with Customized Guidance". In: Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023.	No author or title match. Doesn't exist in publication.
OCTDiff: Bridged Diffusion Model for Portable OCT Super-Resolution and Enhancement	Sources AI	Author Song and AnotherAuthor Zhang. "Consistency in Diffusion Models: Improving Noise Embeddings". In: IEEE Transactions on Pattern Analysis and Machine Intelligence (2023). URL: https://arxiv.org/abs/2304.08787.	No author or title match. This paper has a similar title. ArXiv ID leads to unrelated paper.
Strategic Costs of Perceived Bias in Fair Selection	Sources AI	Claudia Goldin. Occupational choices and the gender wage gap. American Economic Review, 104(5):348-353, 2014.	Author is a famous economist, but the title doesn't match any of her works. Journal and locators match this unrelated article.
Linear Transformers Implicitly Discover Unified Numerical Algorithms	Sources AI	Olah, C., Elhage, N., Nanda, N., Schiefer, N., Jones, A., Henighan, T., and DasSarma, N. (2022). Transformer circuits. Distill, 7(3). https://distill.pub/2022/circuits/.	Most authors match this paper, but the title, publisher, and year are different. Doesn't exist in publication.
Linear Transformers Implicitly Discover Unified Numerical Algorithms	Sources AI	Nanda, N. (2023). Progress in mechanistic interpretability: Reverse-engineering induction heads in GPT-2.	No title match. Author may be Neel Nanda, who wrote several similar articles in 2023.
A Tri-Modal Multi-Agent Responsive Framework for Comprehensive 3D Object Annotation	Sources AI*	J. Zhang and X. Li. Multi-agent systems for distributed problem solving: A framework for task decomposition and coordination. Procedia Computer Science, 55:1131-1138, 2015.	No author or title match. Doesn't exist in publication.
A Tri-Modal Multi-Agent Responsive Framework for Comprehensive 3D Object Annotation	Sources AI*	Erfan Aghasian, Shai Avidan, Piotr Dollar, and Justin Johnson. Hierarchical protocols for multi-agent 3d scene understanding. In CVPR, pages 7664-7673, 2021.	No author or title match. Doesn't exist in publication.
Learning Grouped Lattice Vector Quantizers for Low-Bit LLM Compression	Sources AI*	Itay Hubara, Matthieu Courbariaux, Daniel Soudry, Rami El-Yaniv, and Yoshua Bengio. Quantizing deep convolutional networks for efficient inference: A whitepaper. arXiv preprint arXiv:1612.01462, 2017.	Authors mostly match this paper. Title matches this paper. ArXiv ID matches a third paper.
Learning Grouped Lattice Vector Quantizers for Low-Bit LLM Compression	Sources AI*	Zhiqiang Wang, Chao Zhang, Bing Li, Zhen Xu, and Zhiwei Li. A survey of model compression and acceleration for deep neural networks. ACM Computing Surveys, 54(7):1-34, 2021.	No author match. Title matches this paper. Doesn't exist in publication.
PointMapPolicy: Structured Point Cloud Processing for Multi-Modal Imitation	Sources AI*	Andrew Black et al. Zero-shot skill composition with semantic feature fusion. arXiv preprint arXiv:2310.08573, 2023.	No title match. ArXiv ID leads to unrelated paper.

Learning	AI	arXiv preprint arXiv:2310.06575, 2023.	
PointMapPolicy: Structured Point Cloud Processing for Multi-Modal Imitation Learning	Sources AI*	Yufei Wu, Kiran Alwala, Vivek Ganapathi, Sudeep Sharma, Yilun Chang, Yicheng Zhang, Yilun Zhou, et al. Susie: Scaling up instruction-following policies for robot manipulation. arXiv preprint arXiv:2402.17552, 2024.	No author or title match. ArXiv ID leads to unrelated article.
FLAME: Fast Long-context Adaptive Memory for Event-based Vision	Sources AI	Zhipeng Zhang, Chang Liu, Shihan Wu, and Yan Zhao. EST: Event spatio-temporal transformer for object recognition with event cameras. In ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 1-5. IEEE, 2023.	No author or title match. Doesn't exist in publication.
FLAME: Fast Long-context Adaptive Memory for Event-based Vision	Sources AI	Daniel Gehrig, Mathias Gehrig, John Monaghan, and Davide Scaramuzza. Recurrent vision transformers for dense prediction. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, pages 3139-3148, 2021.	No author match, but this paper has a similar title. Doesn't exist in publication.
Diversity Is All You Need for Contrastive Learning: Spectral Bounds on Gradient Magnitudes	Sources AI**	Qiyang Du, Ozan Sener, and Silvio Savarese. Agree to disagree: Adaptive learning with gradient disagreement. In Advances in Neural Information Processing Systems (NeurIPS), 2021.	No author or title match. Sener and Savarese have published together previously. Doesn't exist in publication.
Diversity Is All You Need for Contrastive Learning: Spectral Bounds on Gradient Magnitudes	Sources AI**	Longxuan Jing, Yu Tian, Yujun Pei, Yibing Shen, and Jiashi Feng. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In International Conference on Learning Representations (ICLR), 2022.	No author match. Title matches this paper. Doesn't exist in publication.
TokenSwap: A Lightweight Method to Disrupt Memorized Sequences in LLMs	Sources AI*	Yair Leviathan, Clemens Rosenbaum, and Slav Petrov. Fast inference from transformers via speculative decoding. In ICML, 2023.	Title, publisher, and date match this paper, but all authors except one surname (Leviathan) are different.
TokenSwap: A Lightweight Method to Disrupt Memorized Sequences in LLMs	Sources AI*	Wenwen Chang, Tal Schuster, and Yann LeCun. Neural surgery for memorisation: Locating and removing verbatim recall neurons. In NeurIPS, 2024.	No author or title match. Doesn't exist in publication.
Learning the Wrong Lessons: Syntactic-Domain Spurious Correlations in Language Models	Sources AI	M. Garcia and A. Thompson. Applications of llms in legal document analysis. Journal of Legal Technology, 7(1):50-65, 2024.	No author or title match. Publication doesn't exist.
Learning the Wrong Lessons: Syntactic-Domain Spurious Correlations in Language Models	Sources AI	J. Smith and A. Patel. Leveraging large language models for financial forecasting. International Journal of Financial Technology, 9(2):101-115, 2024.	No author or title match. Publication doesn't exist.
JADE: Joint Alignment and Deep Embedding for Multi-Slice Spatial Transcriptomics	Sources AI*	David Jones et al. Gpsa: Gene expression and histology-based spatial alignment. Nature Methods, 2023.	No author or title match. Doesn't exist in publication.
JADE: Joint Alignment and Deep Embedding for Multi-Slice Spatial Transcriptomics	Sources AI*	Zhihao Chen, Hantao Zhang, Yuhan Zhang, Zhanlin Hu, Quanquan Gu, Qing Zhang, and Shuo Suo. Slat: a transformer-based method for simultaneous alignment and clustering of spatial transcriptomics data. Nature Communications, 14(1):5548, 2023.	No author or title match. Doesn't exist in publication.
Improved Regret Bounds for Linear Bandits with Heavy-Tailed Rewards	Sources AI	François Baccelli, Gérard H. Taché, and Etienne Altman. Flow complexity and heavytailed delays in packet networks. Performance Evaluation, 49(1-4):427-449, 2002.	No author or title match. Doesn't exist in publication.
Improved Regret Bounds for Linear Bandits with Heavy-Tailed Rewards	Sources AI	Saravanan Jebarajakirthy, Paurav Shukla, and Prashant Palvia. Heavy-tailed distributions in online ad response: A marketing analytics perspective. Journal of Business Research, 124:818-830, 2021.	No author or title match. Doesn't exist in publication.
AutoSciDACT: Automated Scientific Discovery through Contrastive Embedding and Hypothesis Testing	Sources AI	Mehdi Azabou, Micah Weber, Wenlin Ma, et al. Mineclip: Multimodal neural exploration of clip latents for automatic video annotation. arXiv preprint arXiv:2210.02870, 2022.	No author or title match. ArXiv ID leads to unrelated article.

Is there a specific report or published article you think we should check for hallucinations?
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Update: The authors of "AutoSciDACT: Automated Scientific Discovery through Contrastive Embedding and Hypothesis Testing" noted that a mistake was made in the generation of that reference, and the correct reference is:

Mehdi Azabou, Mohammad Gheshlaghi Azar, Ran Liu, Chi-Heng Lin, Erik C Johnson, Kiran Bhaskaran-Nair, Max Dabagia, Bernardo Avila-Pires, Lindsey Kitchell, Keith B Hengen, et al. Mine your own view: Self-supervised learning through across-sample prediction. arXiv preprint arXiv:2102.10106, 2021."

Defining Hallucinated Citations

Given the high stakes for both authors and publishers, GPTZero's Hallucination Check is engineered to be accurate, transparent, and cautious. It uses our AI agent, trained in-house, to flag any citations in a document that can't be found online. These flagged citations are not automatically hallucinations — many archival documents or unpublished works can't be matched to an online source — but they indicate which sources require further human scrutiny. As always, we recommend that a human confirm that flagged citation is an AI-generated fake instead of the result of a more conventional error.

We define a vibe citation as a citation that likely resulted from the use of generative AI. Vibe citing results in errors common to LLM generations, but rare in human-written text, such as:

1. Combining or paraphrasing the titles, author(s), and/or locators from one or more real sources
2. Fabricating the author(s), title, URL/DOI, and/or container (ex. publisher, journal, conference) of a source
3. Modifying the author(s) or title of a source by extrapolating a first name from an initial, dropping and/or adding authors, or paraphrasing the title.

Our definition excludes obvious spelling mistakes, dead URLs, missing locators, and other errors that are plausibly human. The following table shows the difference between a real citation, a flawed citation, and a hallucinated citation according to our methodology. The differences are highlighted in red.

Real Citation	Flawed Citation	Hallucinated Citation
Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. nature, 521:436-444, 2015.	Y. LeCun, Y. Bengio, and Geoff Hinton. Deep leaning. nature, 521(7553):436-444, 2015.	Samuel LeCun Jackson. Deep learning. *Science & Nature*: 23-45, 2021.
A. Yang, B. Zhang, B. Hui, B. Gao, B. Yu, C. Li, D. Liu, J. Tu, J. Zhou, J. Lin, et al. Qwen2.5–math technical report: Toward mathematical expert model via self-improvement. arXiv:2409.12122, 2024.	A. Yang, (missing author), B. Hui, B. Gao, B. Yu, C. Li/, D. Liu, J. Tu, J. Zhou, J. Lin, et al. Qwen 2. 5–math technical report: Toward mathematical expert model via self-improvement. arXiv preprint arXiv:2409.12122, 2024.	A. Yang, B. Yang, C. Yang, et al. Qwen3.5–mathematical report for iterative model self-improvement. arXiv:2909.12233, 2024.

Like GPTZero's AI Detector, Hallucination Check has an extremely low false negative rate, so we catch 99 out of 100

flawed citations. Because our tool will flag any citation that can't be verified online, the false positive rate is higher.

Vibe Citing

Over the past few months, we've experimented with several names for an LLM-generated citation with fabricated elements. "Hallucinated citations" is too long, "hallucitations" too easily mistaken for a spelling error, and "fake citations" too morally charged. Recently, GPTZero's Head of Machine Learning, Alex Adams, coined the term "vibe citing" to describe the LLM tendency to derive or amalgamate real sources into uncanny imitations. "Vibe citing," like "vibe writing" or "vibe coding" produces citations that look accurate at first glance, but crumble under closer inspection.

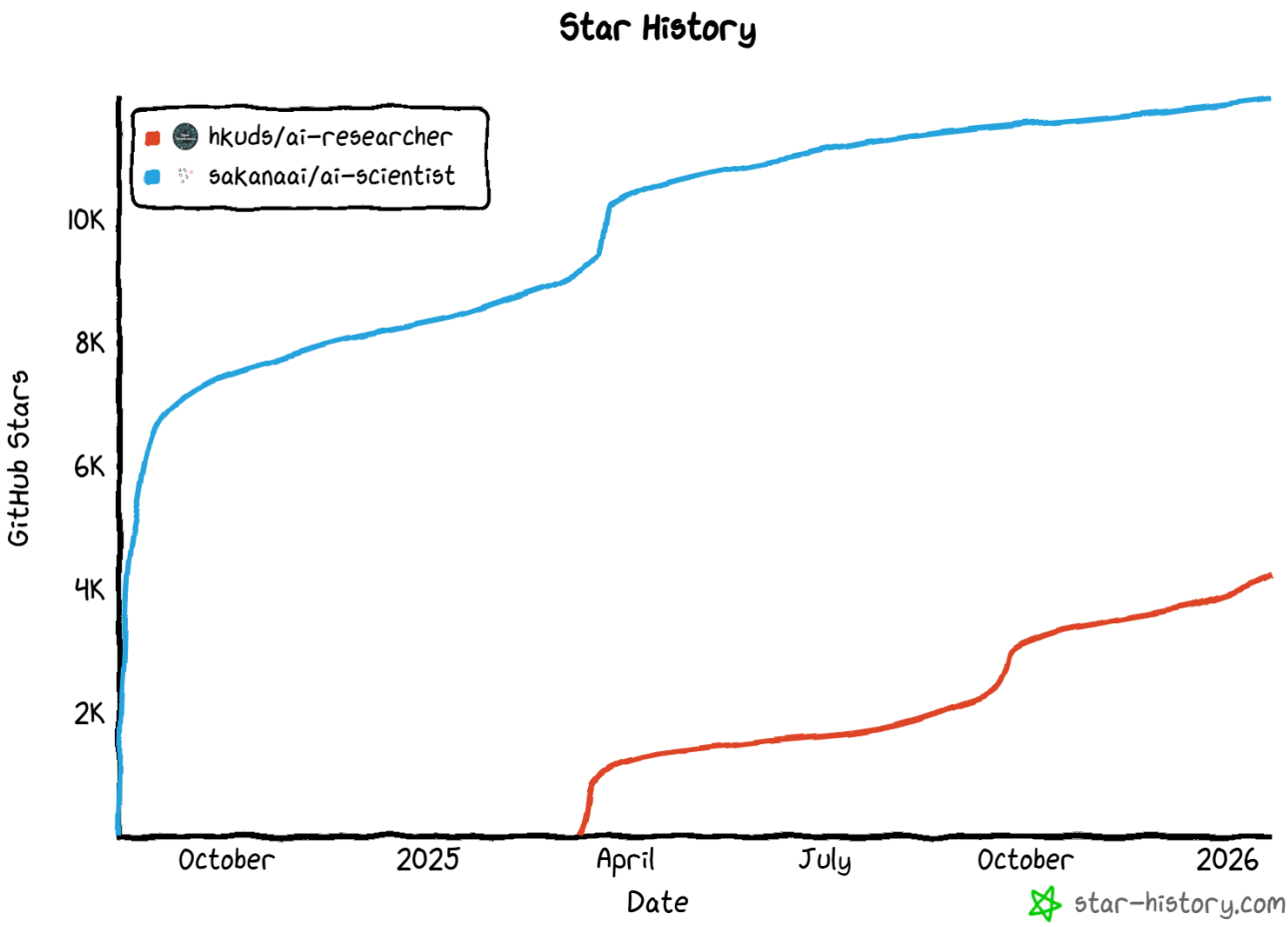


Figure 2: Open-source projects to write research papers with AI are booming in popularity and illustrate the growth in vibe-citing. The bumps in April and September 2025 correspond to the paper submission deadlines for NeurIPS and ICLR 2025.

GPTZero's analysis of 4841 of the 5290 papers accepted by NeurIPS 2025 indicates noticeable traces of AI authorship and hundreds of vibe citations. As always, each of the hallucinations presented here has been verified by a human expert.

Surf the Tsunami with Hallucination Check

Hallucination Check is the only tool of its kind, and provides an essential service at multiple points in the peer review pipeline. First, it allows authors to check their manuscripts for citation errors — including common issues that can occur without LLM involvement like dead links or partial titles. Second, it greatly reduces the time and labor necessary for reviewers to check a submission's sources and identify possible vibe citing. Third, using Hallucination Check in combination with GPTZero's AI Detector allows editors and conference chairs to check for AI-generated text and suspicious citations at the same time, leading to faster and more accurate editorial decisions.

After releasing our ICLR paper investigation we are now coordinating with the ICLR team to review future paper submissions. As always, our goal is to make the peer review process faster, fairer, and more transparent for everyone

involved. *Try GPTZero's Hallucination check for yourself, or reach out to GPTZero's team.*

GPTZero uncovers 50+ Hallucinations in ICLR 2026

GPTZero used our Hallucination Check tool to find 50+ hallucinations under review at ICLR, each of which were missed by 3-5 peer reviewers.

AI Detection Resources | GPTZero
•Paul Esau

The Deloitte Citation Situation - $98K Controversy Explained

GPTZero used our Citation Check to analyze the 234 page report and identified more than 30 issues out of the total 141 citations, including 19 hallucinations. Using GPTZero's citation check would have saved ~$5000 per citation, all within minutes.

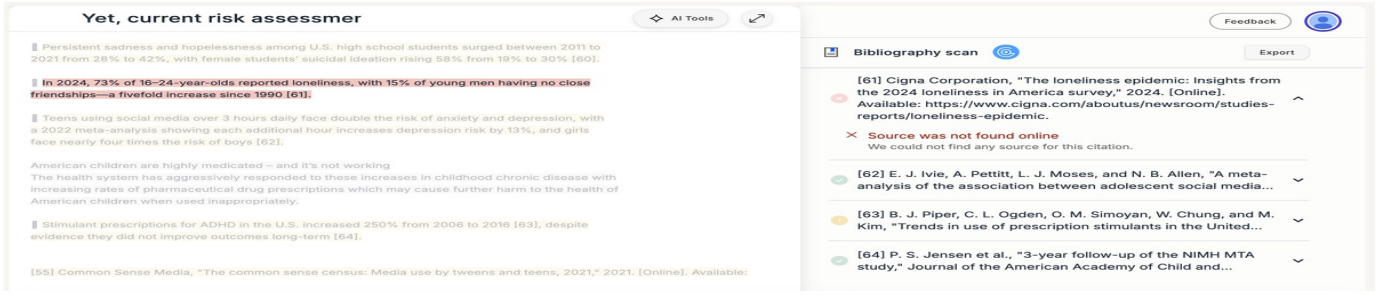
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Making America Hallucinate Again? GPTZero Detects New Errors in Major Government Report

On May 22, the U.S. Presidential Commission to Make America Healthy Again (MAHA), led by health secretary Robert F. Kennedy Jr., released a major report on the causes of chronic diseases in children. Yet within a week, news outlets including NOTUS, the…

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Investigations

Case Study

Research

Written by Nazar Shmatko

KEEP READING

INVESTIGATIONS

GPTZero Team

We Analyzed 134 Moltbook Posts: Human Posts Outperform AI

We found that Moltbook is mainly AI-generated, but the small number of human-written posts drove a disproportionate share of the conversation
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INVESTIGATIONS FEATURED

Paul Esau, Nazar Shmatko, Alex Adam

GPTZero finds over 50 new hallucinations in ICLR 2026 submissions

GPTZero used our Hallucination Check tool to find 50+ hallucinations under review at ICLR, each of which were missed by 3-5 peer reviewers.
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